

MACHINE LEARNING MASTERCLASS

Demystifying Bagging & Boosting

Ensemble Learning Strategies for High-Performance Machine Learning

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What is Ensemble Learning?

Combining weak base learners to construct a highly accurate, robust, and generalizable predictive model.

Core Philosophy

Instead of relying on a single decision model, ensemble methods aggregate predictions from multiple base models to produce a single superior prediction.

Key Concepts

Weak Learner: A model performing slightly better than random guessing (e.g., shallow decision tree).

Strong Learner: An aggregated ensemble achieving high accuracy and generalization.

Error Optimization Goal

Target Total Error Breakdown: ***Total Error = Bias + Variance + Noise***

Bagging vs. Boosting

Quick reference breakdown of core differences in architecture and objective.

Parameter	Bagging	Boosting
Primary Objective	Reduces Variance (Solves Overfitting)	Reduces Bias (Solves Underfitting)
Model Training	Parallel & Independent	Sequential & Adaptive
Data Sampling	Bootstrap Sampling (With Replacement)	Weighted Re-sampling (Focus on Errors)
Aggregation	Simple / Weighted Averaging	Weighted Majority Voting
Classic Examples	Random Forest Classifier	XGBoost, AdaBoost, LightGBM

Bootstrap Aggregating

Bagging reduces variance by averaging predictions across independent bootstrapped trees.

1. Bootstrap Sampling

Generates multiple subset datasets from original training data by sampling **with replacement**. Each subset trains an independent decision tree.

2. Robust Aggregation

Combines outputs from all models to deliver stable predictions:

- **Classification:** Uses Majority Voting.
- **Regression:** Uses Average Prediction score.

The 4-Step Bagging Process

01 Bootstrap Sample

Sample training dataset randomly with replacement given N observations and M features.

02 Feature Selection

Randomly select a subset of m features ($m = \sqrt{M}$) to find optimal splits independently.

03 Tree Growth

Grow each decision tree fully to its maximum depth without pruning.

04 Ensemble Aggregation

Aggregate predictions via majority voting (classification) or mean averaging (regression).

Sequential Error Correction

Boosting builds an ensemble sequentially, training each new learner to fix mistakes made by previous models.

Adaptive Sample Weighting

Misclassified samples receive higher weights in subsequent iterations, forcing the next weak learner to focus on hard-to-predict cases.

Key Advantages & Trade-offs

Pros: Captures complex non-linear feature interactions; achieves exceptional accuracy.

Cons: Sensitive to noisy data & outliers; risk of overfitting if hyper-parameters (e.g. learning rate) aren't tuned.

The 4-Step Boosting Process

01 Initial Weak Learner (d1)

Train initial base decision tree on uniformly weighted sample dataset.

02 Weighted Focus (d2)

Increase weights of misclassified instances and train learner C2 to correct errors.

03 Disagreement Resolution (d3)

Identify samples where preceding learners disagree and train C3 on conflicting regions.

04 Weighted Majority Voting

Combine all weak models into a final strong predictor using weighted majority voting based on individual accuracy.

Random Forest (Bagging) Code

```
# Import Random Forest Classifier
from sklearn.ensemble import RandomForestClassifier

# Initialize Ensemble Model (80 Trees)
rfm = RandomForestClassifier(
    n_estimators=80,
    oob_score=True,
    max_features=0.50,
    min_samples_leaf=5,
    random_state=101,
    n_jobs=-1
)

# Fit Model and Predict Probabilities
rfm.fit(X_train, y_train)
y_pred_prob = rfm.predict_proba(X_test)
```

Summary Takeaway

Choose **Bagging** for high variance / overfitting models. Choose **Boosting** for high bias / underfitting models requiring maximum predictive power.